Linear Algebra
HSLU, Semester 4

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Last update: June 18, 2026

1 Vectors

1.1 Vector arithmetic

Let $\vec{a}, \vec{b}, \vec{c} \in \mathbb{R}^n$ and $\lambda, \mu \in \mathbb{R}$. Then:

$$\begin{aligned} \vec{a} + \vec{b} &= \vec{b} + \vec{a}, & (\vec{a} + \vec{b}) + \vec{c} &= \vec{a} + (\vec{b} + \vec{c}), \\ \vec{a} + \vec{0} &= \vec{a}, & \vec{a} + (-\vec{a}) &= \vec{0}, \\ \lambda(\vec{a} + \vec{b}) &= \lambda\vec{a} + \lambda\vec{b}, & (\lambda + \mu)\vec{a} &= \lambda\vec{a} + \mu\vec{a}, \\ (\lambda\mu)\vec{a} &= \lambda(\mu\vec{a}), & 1\vec{a} &= \vec{a}. \end{aligned}$$

For points $P, Q \in \mathbb{R}^n$, the vector from P to Q is

$$\boxed{\overrightarrow{PQ} = Q - P}$$

Component-wise addition and scalar multiplication are:

$$\begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} + \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix} = \begin{pmatrix} a_1 + b_1 \\ \vdots \\ a_n + b_n \end{pmatrix}, \qquad \lambda \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} = \begin{pmatrix} \lambda a_1 \\ \vdots \\ \lambda a_n \end{pmatrix}.$$

1.2 Length and unit vectors

The Euclidean length of $\vec{a} = (a_1, \dots, a_n)$ is

$$\boxed{\|\vec{a}\|_2 = \sqrt{a_1^2 + \cdots + a_n^2}}$$

If $\vec{a} \neq \vec{0}$, its normalized vector is

$$\hat{\vec{a}} = \frac{\vec{a}}{\|\vec{a}\|_2}.$$

The standard basis vectors of $\mathbb{R}^3$ are and $\|\vec{e}_1\| = \|\vec{e}_2\| = \|\vec{e}_3\| = 1$.

$$\boxed{\vec{e}_1 = \vec{i} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \qquad \vec{e}_2 = \vec{j} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \qquad \vec{e}_3 = \vec{k} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}}$$

$$\boxed{\lambda_1 \vec{v}_1 + \cdots + \lambda_k \vec{v}_k = \sum_{i=1}^{k} \lambda_i \vec{v}_i}$$

$$\boxed{\text{span}(\vec{v}_1, \ldots, \vec{v}_k) = \left\{ \sum_{i=1}^{k} \lambda_i \vec{v}_i \,\middle|\, \lambda_i \in \mathbb{R} \right\}}$$

1.3 Linear combinations and span

A sum of scalar multiples of vectors is called a linear combination:

The span is the set of all such linear combinations:

Every $\vec{x} \in \mathbb{R}^n$ can be expressed using the standard basis:

$$\vec{x} = (x_1, \ldots, x_n) = x_1 \vec{e}_1 + \cdots + x_n \vec{e}_n.$$

1.4 Dot product

For $\vec{a}, \vec{b} \in \mathbb{R}^n$:

$$\boxed{\vec{a} \cdot \vec{b} = \vec{a}^T \vec{b} = \sum_{i=1}^{n} a_i b_i = \|\vec{a}\| \|\vec{b}\| \cos \varphi}$$

The dot product satisfies

$$\begin{aligned} \vec{a} \cdot \vec{a} &\geq 0, & \vec{a} \cdot \vec{a} &= 0 \Longleftrightarrow \vec{a} = \vec{0}, \\ \vec{a} \cdot \vec{b} &= \vec{b} \cdot \vec{a}, & (\vec{a} + \vec{b}) \cdot \vec{c} &= \vec{a} \cdot \vec{c} + \vec{b} \cdot \vec{c}, \\ (\lambda \vec{a}) \cdot \vec{b} &= \lambda (\vec{a} \cdot \vec{b}). \end{aligned}$$

Two vectors are orthogonal exactly when

$$\boxed{\vec{a} \perp \vec{b} \Longleftrightarrow \vec{a} \cdot \vec{b} = 0}$$

For non-zero vectors, the angle between them is

$$\varphi = \arccos\left(\frac{\vec{a} \cdot \vec{b}}{\|\vec{a}\| \|\vec{b}\|} \right).$$

If α, β, γ are the angles between a vector and the x-, y-, and z-axes, then

$$\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1.$$

Mechanical work by a constant force is

$$W = \vec{F} \cdot \vec{s} = \|\vec{F}\| \|\vec{s}\| \cos \varphi.$$

1.5 Cross product (vector product)

For $\vec{u} = (u_1, u_2, u_3)$ and $\vec{v} = (v_1, v_2, v_3)$:

$$\boxed{\vec{u} \times \vec{v} = (u_2 v_3 - u_3 v_2,\ -(u_1 v_3 - u_3 v_1),\ u_1 v_2 - u_2 v_1)}$$

Equivalently, using reduced determinants:

$$\boxed{\vec{u} \times \vec{v} = \left(\begin{vmatrix} u_2 & u_3 \\ v_2 & v_3 \end{vmatrix}, -\begin{vmatrix} u_1 & u_3 \\ v_1 & v_3 \end{vmatrix}, \begin{vmatrix} u_1 & u_2 \\ v_1 & v_2 \end{vmatrix} \right)}$$

The cross product is perpendicular to both vectors and

$$\|\vec{u} \times \vec{v}\| = \|\vec{u}\| \|\vec{v}\| \sin\varphi$$

is the area of the parallelogram they span. Furthermore,

$$\begin{aligned} \vec{u} \times \vec{v} &= -(\vec{v} \times \vec{u}), \\ \vec{u} \times (\vec{v} + \vec{w}) &= \vec{u} \times \vec{v} + \vec{u} \times \vec{w}, \\ (\lambda \vec{u}) \times \vec{v} &= \lambda(\vec{u} \times \vec{v}), \\ \vec{u} \times \vec{v} &= \vec{0} \iff \vec{u}, \vec{v} \quad \text{are collinear (or one is zero).} \end{aligned}$$

2 Matrices

2.1 Definitions and special matrices

An $m \times n$ matrix has m rows and n columns:

$$A = (a_{ij}) \in \mathbb{R}^{m \times n}.$$

A matrix is square if $m = n$. A matrix in $\mathbb{R}^{n \times 1}$ is a column vector, and one in $\mathbb{R}^{1 \times n}$ is a row vector.

Important special matrices are:

$$O = (0)_{m \times n}, \qquad I_n = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix},$$

upper and lower triangular matrices, and diagonal matrices $D = \text{diag}(d_1, \ldots, d_n)$.

The transpose interchanges rows and columns:

$$(A^T)_{ij} = a_{ji}.$$

A square matrix is symmetric if

$$\boxed{A^T = A}$$

2.2 Addition and scalar multiplication

For matrices of equal size:

$$(A + B)_{ij} = a_{ij} + b_{ij}, \qquad (\lambda A)_{ij} = \lambda a_{ij}.$$

The usual associative, commutative, distributive, zero, and additive inverse laws apply.

2.3 Matrix multiplication

If $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{n \times p}$, then $AB \in \mathbb{R}^{m \times p}$ with

$$\boxed{(AB)_{ij} = \sum_{k=1}^{n} a_{ik} b_{kj}}$$

The multiplication rules are

$$\begin{aligned} (AB)C &= A(BC), & (A + B)C &= AC + BC, \\ A(B + C) &= AB + AC, & \lambda(AB) &= (\lambda A)B = A(\lambda B), \\ AI &= IA = A, & (AB)^T &= B^T A^T. \end{aligned}$$

In general, $AB \neq BA$.

3 Linear Systems and Gaussian Elimination

A linear system can be written as where A is the coefficient matrix, $\vec{x}$ is the vector of unknowns, and $\vec{b}$ is

$$\boxed{A\vec{x} = \vec{b}}$$

the right-hand side. Its augmented matrix is $[A \mid \vec{b}]$.

3.1 Elementary row operations

Gaussian elimination uses:

1. interchange two rows;
2. multiply a row by a non-zero scalar;
3. add a multiple of one row to another row.

The objective is row echelon form, with each pivot farther right than the pivot above it and zero entries below every pivot:

$$\left[\begin{array}{ccc|c} * & * & * & * \\ 0 & * & * & * \\ 0 & 0 & * & * \end{array}\right].$$

Gauss–Jordan elimination additionally produces zeros above each pivot, giving reduced row echelon form.

3.2 Solution cases

- A pivot in every variable column and no contradiction: one unique solution.
- At least one free variable and no contradiction: infinitely many solutions.
- A row $[0 \ \cdots \ 0 \mid c]$ with $c \neq 0$: no solution.

3.3 Homogeneous and inhomogeneous systems

The homogeneous system

$$A\vec{x} = \vec{0}$$

always has the trivial solution $\vec{x} = \vec{0}$. If free variables exist, it also has infinitely many non-trivial solutions.

For a consistent inhomogeneous system $A\vec{x} = \vec{b}$, every solution has the form

$$\boxed{\vec{x} = \vec{x}_p + \vec{x}_h, \qquad A\vec{x}_p = \vec{b}, \quad A\vec{x}_h = \vec{0}}$$

4 Determinants and Inverse Matrices

4.1 Determinant

For a 2×2 matrix,

$$\boxed{\det\begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc}$$

For $A \in \mathbb{R}^{3\times 3}$, Sarrus' rule gives:

$$\left[\begin{array}{ccc|cc} a_{11} & a_{12} & a_{13} & a_{11} & a_{12} \\ a_{21} & a_{22} & a_{23} & a_{21} & a_{22} \\ a_{31} & a_{32} & a_{33} & a_{31} & a_{32} \end{array}\right]$$

$$\det A = a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} - a_{13}a_{22}a_{31} - a_{11}a_{23}a_{32} - a_{12}a_{21}a_{33}.$$

For a triangular matrix,

$$\det A = \prod_{i=1}^{n} a_{ii}.$$

For a general square matrix, reduce it to triangular form while tracking:

- swapping two rows changes the sign of the determinant;
- multiplying a row by λ multiplies the determinant by λ;
- adding a multiple of one row to another does not change the determinant.

Important rules are

$$\det(A^T) = \det A, \qquad \det(\lambda A) = \lambda^n \det A,$$
$$\det(AB) = \det A \det B, \qquad \det(A^{-1}) = \frac{1}{\det A}.$$

A square matrix is singular exactly when $\det A = 0$.

4.2 Inverse matrix

A square matrix A is invertible if there is a matrix A^{-1} such that

$$\boxed{AA^{-1} = A^{-1}A = I_n}$$

For a 2×2 matrix:

$$\boxed{\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad - bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}, \qquad ad - bc \neq 0}$$

In general,

$$A^{-1} = \frac{1}{\det A}\operatorname{adj}(A), \qquad \det A \neq 0.$$

It can also be computed by Gauss–Jordan elimination:

$$[A \mid I_n] \sim [I_n \mid A^{-1}].$$

If A is invertible, the unique solution of $A\vec{x} = \vec{b}$ is

$$\boxed{\vec{x} = A^{-1}\vec{b}}$$

5 Vector Spaces, Subspaces, and Bases

5.1 Subspaces

A subset $U \subseteq \mathbb{R}^n$ is a vector subspace if:

1. $\vec{0} \in U$;
2. $\vec{u}, \vec{v} \in U \implies \vec{u} + \vec{v} \in U$;
3. $\vec{u} \in U,\ \lambda \in \mathbb{R} \implies \lambda\vec{u} \in U$.

The span of any set of vectors is a subspace. In particular, $\operatorname{span}(\vec{v}_1, \dots, \vec{v}_k)$ is the smallest subspace containing all of those vectors.

5.2 Linear independence

Vectors $\vec{v}_1, \dots, \vec{v}_k$ are linearly independent if Otherwise they are linearly dependent.

$$\boxed{\lambda_1 \vec{v}_1 + \cdots + \lambda_k \vec{v}_k = \vec{0} \Longrightarrow \lambda_1 = \cdots = \lambda_k = 0}$$

Two non-zero vectors are dependent exactly when one is a scalar multiple of the other. A single vector is independent exactly when it is non-zero.

If the vectors are columns of a matrix A, they are independent exactly when the homogeneous system $A\vec{x} = \vec{0}$ has only the trivial solution.

5.3 Basis and dimension

The vectors $\vec{v}_1, \dots, \vec{v}_k \in U$ form a basis of U if they:

1. span U;
2. are linearly independent.

Every vector in U then has a unique representation as a linear combination of the basis vectors.

The dimension $\dim U$ is the number of vectors in any basis of U. In particular,

$$\dim \mathbb{R}^n = n, \qquad \dim\{\vec{0}\} = 0.$$

If $\dim U = d$ and $\vec{v}_1, \dots, \vec{v}_d \in U$, then the following are equivalent:

$$\{\vec{v}_1, \dots, \vec{v}_d\} \text{ is a basis of } U \Longleftrightarrow \begin{cases} \vec{v}_1, \dots, \vec{v}_d & \text{are independent,} \\ U = \operatorname{span}(\vec{v}_1, \dots, \vec{v}_d). \end{cases}$$

With exactly d vectors, it is enough to verify either independence or spanning.

5.4 Affine subspaces

An affine subspace is a translated vector subspace: where $\vec{v} \in \mathbb{R}^n$ and $U \subseteq \mathbb{R}^n$ is a subspace. Its dimension

$$\boxed{A = \vec{v} + U = \{\vec{v} + \vec{u} \mid \vec{u} \in U\}}$$

is $\dim A = \dim U$.

$\dim A$	geometric object
0	point
1	line
2	plane
$n-1$	hyperplane

A consistent solution set of $A\vec{x} = \vec{b}$ is an affine subspace: $\vec{x}_p + \ker A$.

6 Linear and Affine Maps

6.1 Linear maps

A map $L : \mathbb{R}^n \to \mathbb{R}^m$ is linear if, for all $\vec{u}, \vec{v} \in \mathbb{R}^n$ and $\lambda, \mu \in \mathbb{R}$, Equivalently, it is additive and homogeneous:

$$\boxed{L(\lambda \vec{u} + \mu \vec{v}) = \lambda L(\vec{u}) + \mu L(\vec{v})}$$

$$L(\vec{u} + \vec{v}) = L(\vec{u}) + L(\vec{v}), \qquad L(\lambda \vec{u}) = \lambda L(\vec{u}).$$

Every linear map satisfies $L(\vec{0}) = \vec{0}$.

Every linear map $L : \mathbb{R}^n \to \mathbb{R}^m$ has a matrix $A \in \mathbb{R}^{m \times n}$ such that

$$L(\vec{x}) = A\vec{x}.$$

In the standard bases, its columns are

$$\boxed{A = \begin{bmatrix} L(\vec{e}_1) & \cdots & L(\vec{e}_n) \end{bmatrix}}$$

6.2 Common transformations

Counter-clockwise rotation through angle θ:

$$R_\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}.$$

Clockwise rotation through angle θ:

$$R_{-\theta} = \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix}.$$

6.3 Affine maps

A map $F : \mathbb{R}^n \to \mathbb{R}^m$ is affine if for a linear map L, a matrix A, and a translation vector $\vec{v}$. It is linear exactly

$$\boxed{F(\vec{x}) = L(\vec{x}) + \vec{v} = A\vec{x} + \vec{v}}$$

when $\vec{v} = \vec{0}$.

6.4 Inverse maps

A linear map $L : \mathbb{R}^n \to \mathbb{R}^n$ is invertible if there is a map L^{-1} such that

$$L^{-1}(L(\vec{x})) = \vec{x}, \qquad L(L^{-1}(\vec{y})) = \vec{y}.$$

If $L(\vec{x}) = A\vec{x}$, then and L is invertible exactly when $\det A \neq 0$.

$$\boxed{L^{-1}(\vec{x}) = A^{-1}\vec{x}}$$

7 Kernel, Image, Rank, and Nullity

For $L(\vec{x}) = A\vec{x}$:

$$\boxed{\ker L = \{\vec{x} \in \mathbb{R}^n \mid A\vec{x} = \vec{0}\}, \qquad \operatorname{im} L = \{A\vec{x} \mid \vec{x} \in \mathbb{R}^n\}}$$

If $A = [\vec{a}_1 \ \cdots \ \vec{a}_n]$, then

$$\operatorname{im} A = \operatorname{span}(\vec{a}_1, \ldots, \vec{a}_n).$$

To obtain a basis of the image, row-reduce A and select the original columns corresponding to pivot columns.

The rank and nullity are

$$\operatorname{rank} A = \dim(\operatorname{im} A), \qquad \operatorname{nullity} A = \dim(\ker A).$$

The rank is the number of pivots; the nullity is the number of free variables. For $A \in \mathbb{R}^{m\times n}$:

$$\boxed{\operatorname{rank} A + \operatorname{nullity} A = n}$$

7.1 Consequences for systems

For $A \in \mathbb{R}^{m\times n}$:

$$\begin{aligned} \ker A = \{\vec{0}\} &\iff \operatorname{nullity} A = 0 \\ &\iff \operatorname{rank} A = n \\ &\iff \text{the columns of } A \text{ are independent.} \end{aligned}$$

Furthermore,

$$\operatorname{im} A = \mathbb{R}^m \iff \operatorname{rank} A = m \iff A\vec{x} = \vec{b} \text{ is solvable for every } \vec{b} \in \mathbb{R}^m.$$

7.2 Equivalent conditions for square matrices

For $A \in \mathbb{R}^{n\times n}$, the following are equivalent:

1. A is invertible;
2. $\det A \neq 0$;
3. $\ker A = \{\vec{0}\}$;
4. $\operatorname{rank} A = n$;
5. $\operatorname{im} A = \mathbb{R}^n$;
6. the rows and columns of A each form a basis of $\mathbb{R}^n$;
7. $A\vec{x} = \vec{0}$ has only the trivial solution;
8. $A\vec{x} = \vec{b}$ has the unique solution $\vec{x} = A^{-1}\vec{b}$ for every $\vec{b} \in \mathbb{R}^n$.

8 Eigenvalues and Eigenvectors

Let $A \in \mathbb{R}^{n\times n}$. A scalar λ is an eigenvalue of A if there is a non-zero vector $\vec{v}$ such that The vector $\vec{v}$ is an

$$\boxed{A\vec{v} = \lambda\vec{v}, \qquad \vec{v} \neq \vec{0}}$$

eigenvector corresponding to λ. Every non-zero scalar multiple of an eigenvector is another eigenvector for the same eigenvalue.

8.1 Eigenspaces

The eigenspace associated with λ is It contains the zero vector; its non-zero elements are the eigenvec-

$$\boxed{E_\lambda = \ker(A - \lambda I) = \{\vec{v} \in \mathbb{R}^n \mid A\vec{v} = \lambda\vec{v}\}}$$

tors associated with λ.

8.2 Computing eigenvalues and eigenvectors

1. Form the characteristic polynomial
$$p_A(\lambda) = \det(A - \lambda I).$$
2. Solve $p_A(\lambda) = 0$. Its roots are the eigenvalues.
3. For each eigenvalue λ_i, solve
$$(A - \lambda_i I)\vec{v} = \vec{0}$$
to find a basis of E_{λ_i}.

9 Diagonalization

A square matrix A is diagonalizable if there is a basis of $\mathbb{R}^n$ consisting of eigenvectors of A. If $\vec{v}_1, \dots, \vec{v}_n$ are independent eigenvectors with eigenvalues $\lambda_1, \dots, \lambda_n$, define

$$S = \begin{bmatrix} \vec{v}_1 & \cdots & \vec{v}_n \end{bmatrix}, \qquad D = \operatorname{diag}(\lambda_1, \dots, \lambda_n).$$

Then

$$\boxed{S^{-1}AS = D \qquad \Longleftrightarrow \qquad A = SDS^{-1}}$$

An $n \times n$ matrix is diagonalizable exactly when the sum of the dimensions of its eigenspaces is n. In particular, n distinct eigenvalues guarantee diagonalizability.

Diagonalization simplifies powers:
$$A^k = SD^kS^{-1}.$$

10 Norms, Metrics, and Orthogonality

10.1 Norms

A norm satisfies, for all vectors $\vec{x}, \vec{y}$ and scalars λ:

$$\begin{aligned} \|\vec{x}\| &\geq 0, \\ \|\vec{x}\| = 0 &\iff \vec{x} = \vec{0}, \\ \|\lambda\vec{x}\| &= |\lambda|\|\vec{x}\|, \\ \|\vec{x} + \vec{y}\| &\leq \|\vec{x}\| + \|\vec{y}\|. \end{aligned}$$

Common norms are

$$\|\vec{x}\|_2 = \sqrt{\sum_{i=1}^{n} x_i^2}, \qquad \|\vec{x}\|_\infty = \max_{1\leq i\leq n} |x_i|.$$

10.2 Metrics

A metric $d : M \times M \to \mathbb{R}$ satisfies:

$$\begin{aligned} d(x,y) &\geq 0, \\ d(x,y) = 0 &\iff x = y, \\ d(x,y) &= d(y,x), \\ d(x,y) &\leq d(x,z) + d(z,y). \end{aligned}$$

A norm induces the metric $d(\vec{x}, \vec{y}) = \|\vec{x} - \vec{y}\|$.

10.3 Orthogonal and orthonormal bases

A basis $\vec{b}_1, \dots, \vec{b}_k$ is orthogonal if

$$\vec{b}_i \cdot \vec{b}_j = 0 \qquad (i \neq j).$$

It is orthonormal if additionally

$$\|\vec{b}_i\| = 1.$$

For an orthonormal basis, the coordinates of $\vec{v}$ are particularly simple:

$$\vec{v} = \sum_{i=1}^{k} (\vec{v} \cdot \vec{b}_i)\vec{b}_i.$$

11 Orthogonal Projections and Gram–Schmidt

11.1 Projection onto a line

The orthogonal projection of $\vec{v}$ onto $L = \text{span}(\vec{b})$, $\vec{b} \neq \vec{0}$, is If $\|\vec{b}\| = 1$, then

$$\boxed{\text{proj}_{\vec{b}}(\vec{v}) = \frac{\vec{v} \cdot \vec{b}}{\vec{b} \cdot \vec{b}}\,\vec{b}}$$

$$\text{proj}_{\vec{b}}(\vec{v}) = (\vec{v} \cdot \vec{b})\vec{b}.$$

The error vector is orthogonal to the line:

$$\vec{v} - \text{proj}_{\vec{b}}(\vec{v}) \perp \vec{b}.$$

11.2 Projection onto a subspace

If $\vec{b}_1, \dots, \vec{b}_k$ is an orthonormal basis of U, then The residual $\vec{v} - \text{proj}_U(\vec{v})$ lies in $U^\perp$.

$$\boxed{\operatorname{proj}_U(\vec{v}) = \sum_{i=1}^{k} (\vec{v}\cdot\vec{b}_i)\vec{b}_i}$$

11.3 Gram–Schmidt process

Starting from a basis $\vec{v}_1, \dots, \vec{v}_k$, construct an orthogonal basis:

$$\begin{aligned}
\vec{u}_1 &= \vec{v}_1, \\
\vec{u}_2 &= \vec{v}_2 - \operatorname{proj}_{\vec{u}_1}(\vec{v}_2), \\
\vec{u}_j &= \vec{v}_j - \sum_{i=1}^{j-1} \operatorname{proj}_{\vec{u}_i}(\vec{v}_j).
\end{aligned}$$

Normalize the resulting vectors: Then $\vec{b}_1, \dots, \vec{b}_k$ is an orthonormal basis of the same subspace.

$$\boxed{\vec{b}_i = \frac{\vec{u}_i}{\|\vec{u}_i\|}, \qquad i = 1, \dots, k}$$

11.4 Algorithm in a nutshell

Step 1. $w_1 \longrightarrow v_1 = \dfrac{w_1}{\|w_1\|}$

Step 2. $v_2' = w_2 - \underbrace{(w_1 \cdot v_1)\, v_1}_{P_{\operatorname{span}(v_1)}(w_2)}$

Step 3. $v_2 = \dfrac{v_2'}{\|v_2'\|}$

Step 4. $v_3' = w_3 - \underbrace{(w_3 \cdot v_1)\, v_1 - (w_3 \cdot v_2)\, v_2}_{P_{\operatorname{span}(v_1, v_2)}(w_3)}$

Step 5. $v_3 = \dfrac{v_3'}{\|v_3'\|}$

12 Spectral Theorem

Every real symmetric matrix $A = A^T$:

- has only real eigenvalues;
- has mutually orthogonal eigenspaces for distinct eigenvalues;
- has an orthonormal basis of eigenvectors;
- is orthogonally diagonalizable.

If the orthonormal eigenvectors are the columns of V and their eigenvalues form D, then $V^{-1} = V^T$ and

$$\boxed{V^T A V = D \qquad \Longleftrightarrow \qquad A = VDV^T}$$